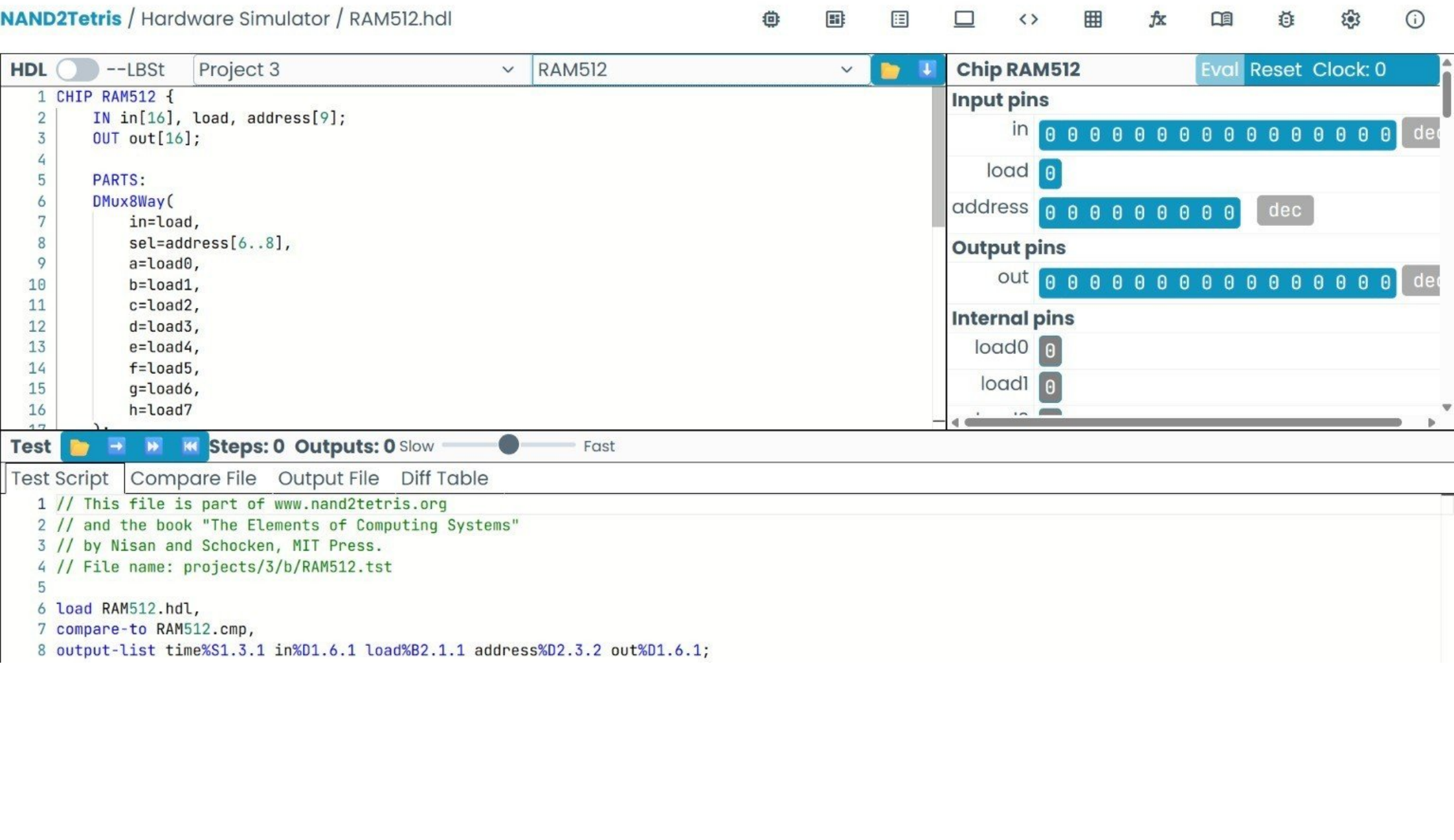
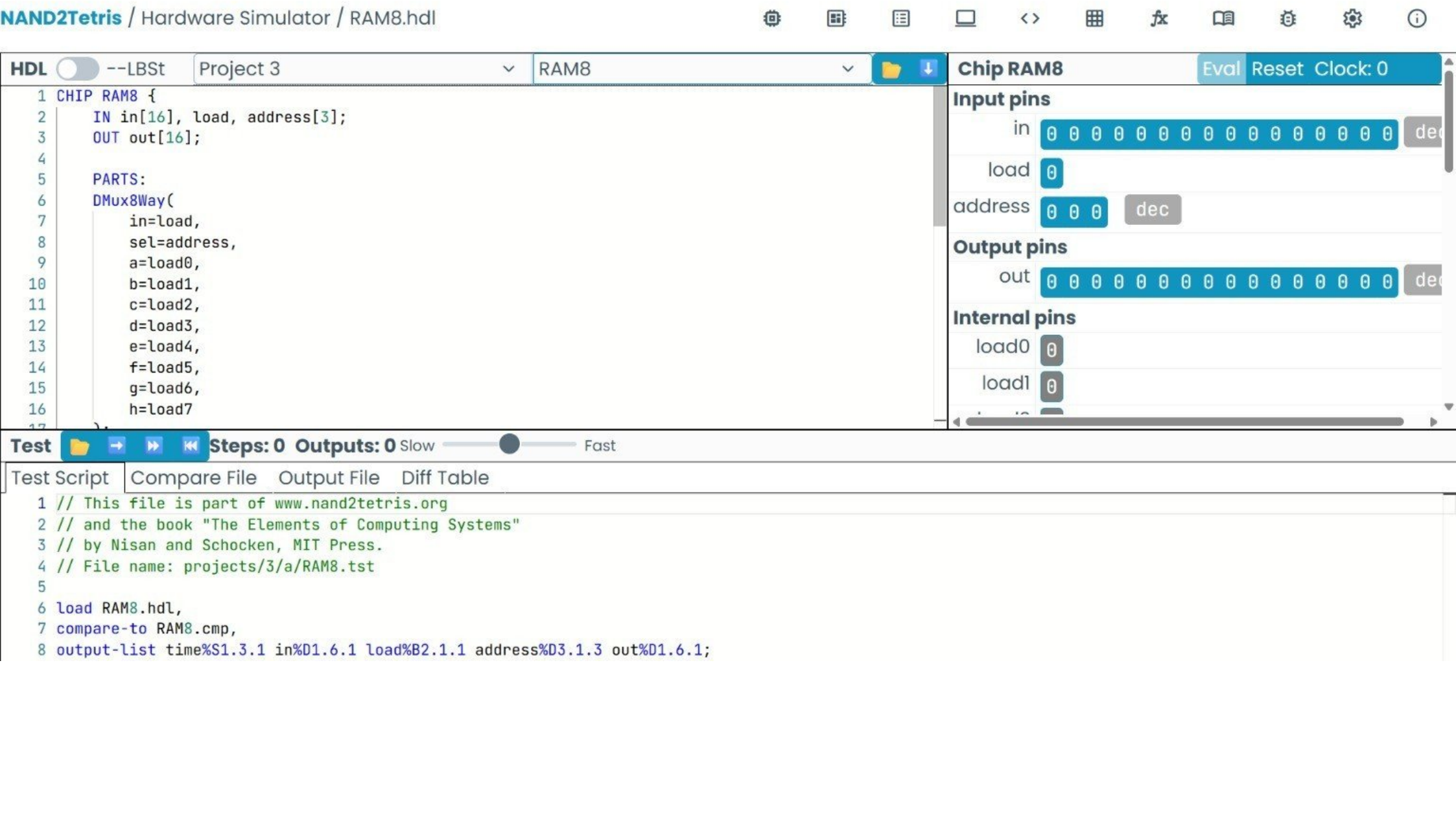

NAND2Tetris / Hardware Simulator / RAM512.hdl
HDL --LBSt Project 3 RAM512
CHIP RAM512 {
IN in[16], load, address[9];
OUT out[16];
PARTS:
DMux8Way(
in=load,
sel=address[6..8],
a=load0,
b=load1,
c=load2,
d=load3,
e=load4,
f=load5,
g=load6,
h=load7
Chip RAM512 Eval Reset Clock: 0
Input pins
in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
load 0
address 0 0 0 0 0 0 0 0 0 dec
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Internal pins
load0 0
load1 0
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.tst
load RAM512.hdl,
compare-to RAM512.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.3.2 out%D1.6.1;


NAND2Tetris / Hardware Simulator / RAM8.hdl
HDL --LBSt Project 3 RAM8
CHIP RAM8 {
IN in[16], load, address[3];
OUT out[16];
PARTS:
DMux8Way(
in=load,
sel=address,
a=load0,
b=load1,
c=load2,
d=load3,
e=load4,
f=load5,
g=load6,
h=load7
Chip RAM8 Eval Reset Clock: 0
Input pins
in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
load 0
address 0 0 0 dec
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Internal pins
load0 0
load1 0
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.tst
load RAM8.hdl,
compare-to RAM8.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D3.1.3 out%D1.6.1;

HDL --LBSt Project 1 Mux16

```
1  CHIP Mux16 {
2      IN a[16], b[16], sel;
3      OUT out[16];
4
5      PARTS:
6      Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
7      Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
8      Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
9      Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
10     Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
11     Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
12     Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
13     Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
14     Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
15     Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
16     Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
```

Chip Mux16 Eval Reset Clock: 0

Input pins

a	1 0 1 0 0 1 0 0 0 1 0 1 0 1 1 0	dec
b	0 0 1 0 1 0 1 1 0 0 0 1 0 0 1 0	dec
sel	1	

Output pins

out	0 0 1 0 1 0 1 1 0 0 0 1 0 0 1 0	dec

Test **Steps: 1 Outputs: 0** Slow Fast

Test Script | Compare File | Output File | Diff Table

HDL code: No syntax errors

NAND2Tetris
/ Hardware Simulator / Not16.hdl

memory developer_board list_alt computer code grid_on function menu_book bug_report settings

HDL --LBSt Project 1 Not16

```
1  CHIP Not16 {
2      IN in[16];
3      OUT out[16];
4
5      PARTS:
6      Not(in=in[0], out=out[0]);
7      Not(in=in[1], out=out[1]);
8      Not(in=in[2], out=out[2]);
9      Not(in=in[3], out=out[3]);
10     Not(in=in[4], out=out[4]);
11     Not(in=in[5], out=out[5]);
12     Not(in=in[6], out=out[6]);
13     Not(in=in[7], out=out[7]);
14     Not(in=in[8], out=out[8]);
15     Not(in=in[9], out=out[9]);
16     Not(in=in[10], out=out[10]);
```

Chip Not16 Eval Reset Clock: 0

Input pins

in 0 1 0 0 1 0 0 1 1 0 0 1 0 0 1 0 dec

Output pins

out 1 0 1 1 0 1 1 0 0 1 1 0 1 1 0 1 dec

Test **Steps: 1 Outputs: 0** Slow Fast

Test Script | Compare File | Output File | Diff Table

HDL code: No syntax errors

HDL --LBSt Project 1 Or

```
1
2  CHIP Or {
3      IN a, b;
4      OUT out;
5
6      PARTS:
7      Nand(a=a,b=a,out=nand1out);
8      Nand(a=b,b=b,out=nand2out);
9      Nand(a=nand1out,b=nand2out,out=out);
10 }
```

Input pins

a	0
b	1

Output pins

out	1

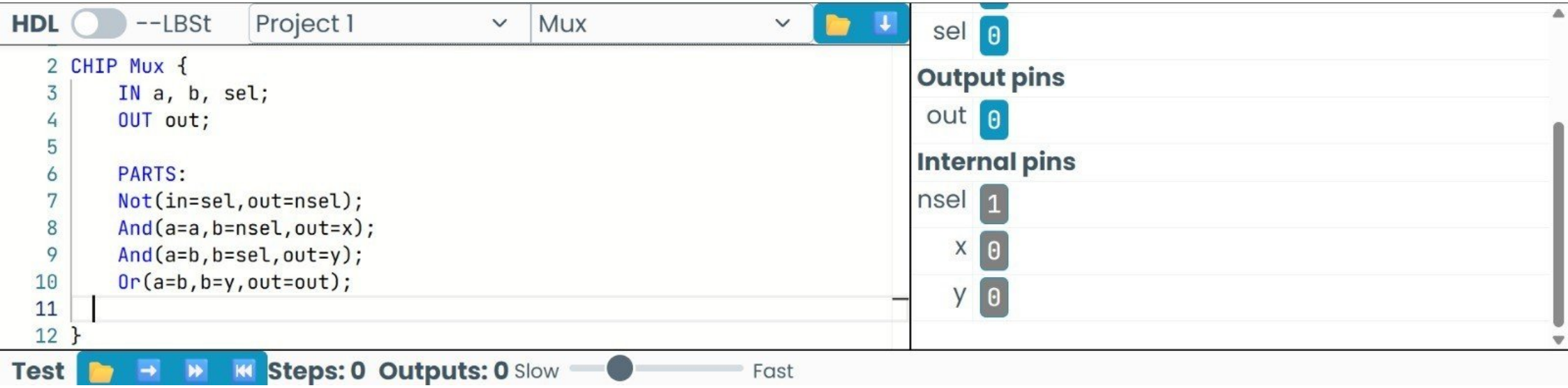
Internal pins

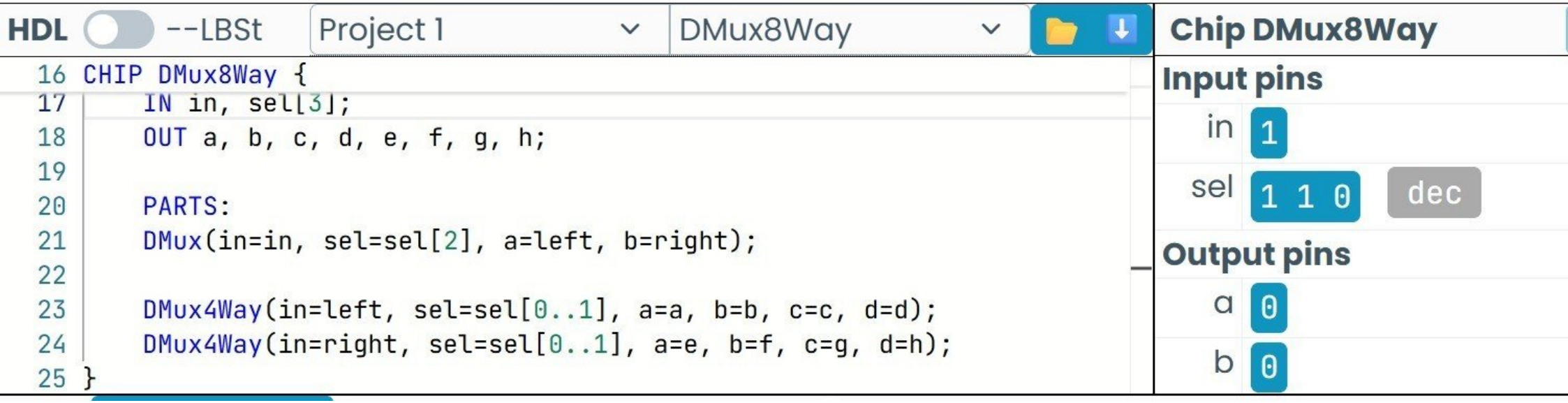
nand1out	1
nand2out	0

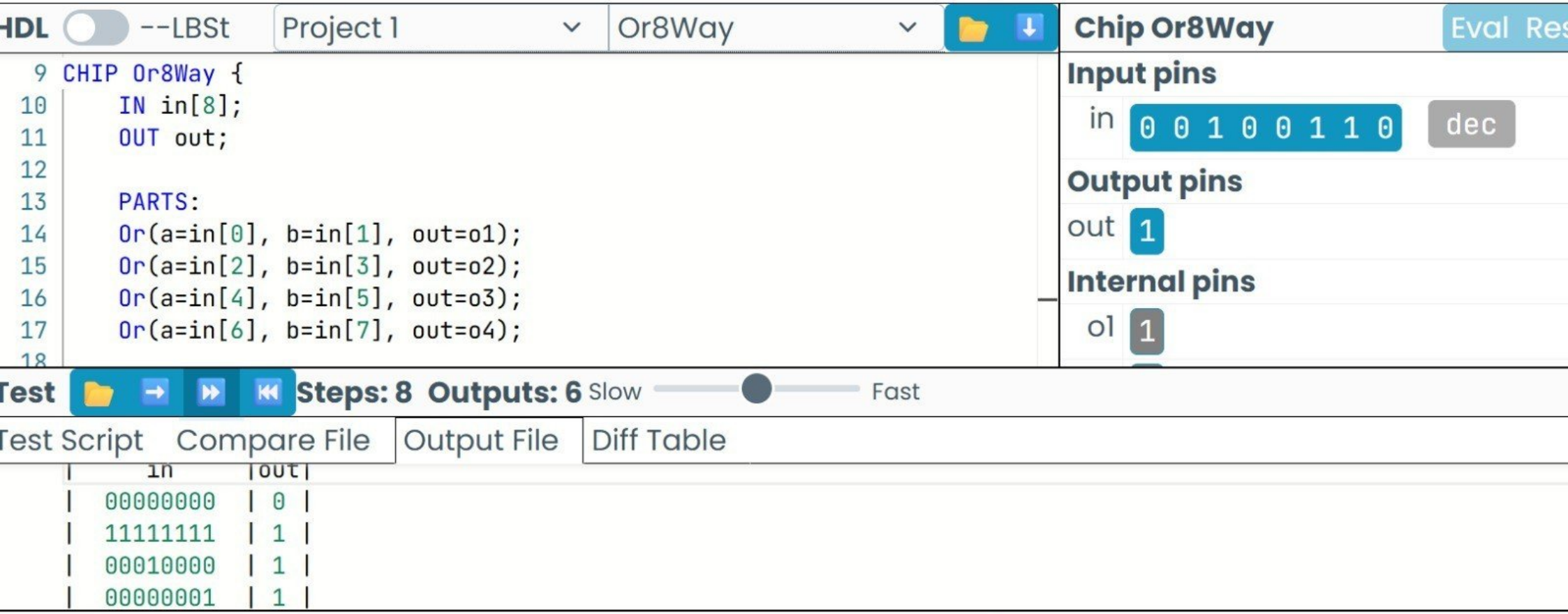
Test **Steps: 0 Outputs: 0** Slow Fast

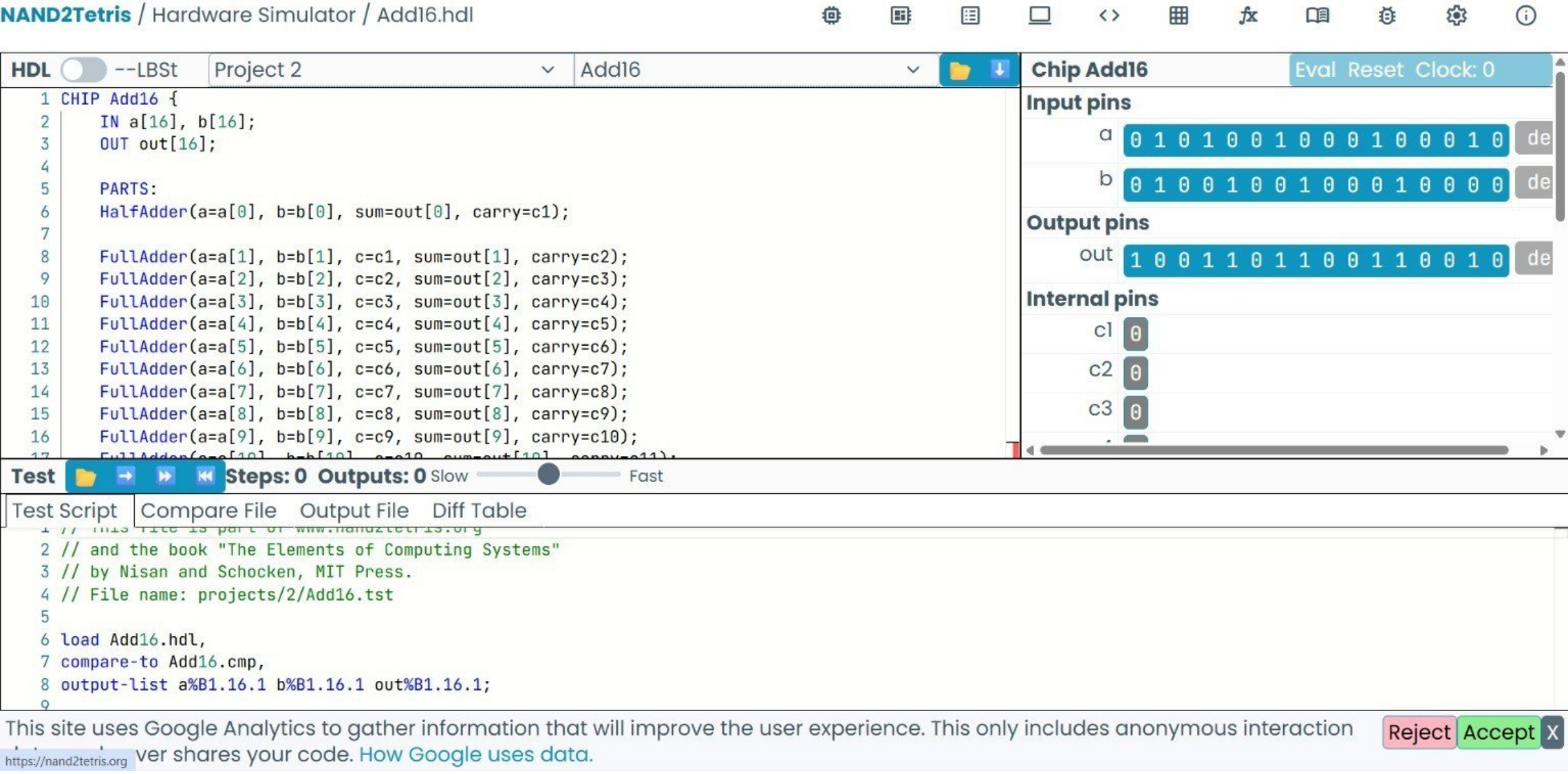
Test Script | Compare File | Output File | Diff Table

```
1  // This file is part of www.nand2tetris.org
2  // and the book "The Elements of Computing Systems"
3  // by Nisan and Schocken, MIT Press.
4  // File name: projects/1/Or.tst
```


HDL
--LBSt
Project 1
Mux
2 CHIP Mux {
3 IN a, b, sel;
4 OUT out;
5
6 PARTS:
7 Not(in=sel,out=nsel);
8 And(a=a,b=nsel,out=x);
9 And(a=b,b=sel,out=y);
10 Or(a=b,b=y,out=out);
11
12 }
sel 0
Output pins
out 0
Internal pins
nsel 1
x 0
y 0
Test
Steps: 0 Outputs: 0 Slow
Fast


HDL
--LBSt
Project 1
DMux8Way
16 CHIP DMux8Way {
17 IN in, sel[3];
18 OUT a, b, c, d, e, f, g, h;
19
20 PARTS:
21 DMux(in=in, sel=sel[2], a=left, b=right);
22
23 DMux4Way(in=left, sel=sel[0..1], a=a, b=b, c=c, d=d);
24 DMux4Way(in=right, sel=sel[0..1], a=e, b=f, c=g, d=h);
25 }
Chip DMux8Way
Input pins
in 1
sel 1 1 0 dec
Output pins
a 0
b 0


HDL
--LBSt
Project 1
Or8Way
9 CHIP Or8Way {
10 IN in[8];
11 OUT out;
12
13 PARTS:
14 Or(a=in[0], b=in[1], out=o1);
15 Or(a=in[2], b=in[3], out=o2);
16 Or(a=in[4], b=in[5], out=o3);
17 Or(a=in[6], b=in[7], out=o4);
Chip Or8Way
Eval Res
Input pins
in
0 0 1 0 0 1 1 0
dec
Output pins
out
1
Internal pins
o1
1
Test
Steps: 8 Outputs: 6
Slow
Fast
Test Script
Compare File
Output File
Diff Table
in	out
00000000	0
11111111	1
00010000	1
00000001	1


NAND2Tetris / Hardware Simulator / Add16.hdl
HDL --LBSt
Project 2
Add16
CHIP Add16 {
IN a[16], b[16];
OUT out[16];
PARTS:
HalfAdder(a=a[0], b=b[0], sum=out[0], carry=c1);
FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry=c2);
FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry=c3);
FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry=c4);
FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry=c5);
FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=c6);
FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry=c7);
FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
Chip Add16
Eval Reset Clock: 0
Input pins
a 0 1 0 1 0 0 1 0 0 0 1 0 0 0 1 0
b 0 1 0 0 1 0 0 1 0 0 0 1 0 0 0 0
Output pins
out 1 0 0 1 1 0 1 1 0 0 1 1 0 0 1 0
Internal pins
c1 0
c2 0
c3 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Add16.tst
load Add16.hdl,
compare-to Add16.cmp,
output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
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https://nand2tetris.org

HDL --LBSt Project 1 Not

```
1
2 CHIP Not {
3     IN in;
4     OUT out;
5
6     PARTS:
7     Nand(a=in,b=in,out=out);
8 }
```

Chip Not Eval Reset Clock: 0

Input pins

in 1

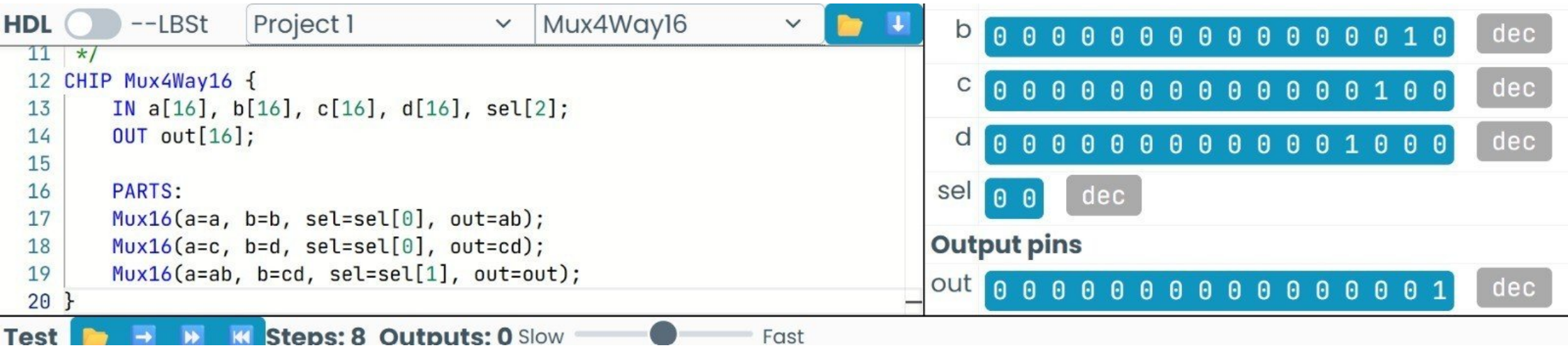
Output pins

out 0

Test Steps: 0 Outputs: 0 Slow Fast

Test Script | Compare File | Output File | Diff Table

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.tst
5
```


HDL
--LBSt
Project 1
Mux4Way16
*/
CHIP Mux4Way16 {
IN a[16], b[16], c[16], d[16], sel[2];
OUT out[16];
PARTS:
Mux16(a=a, b=b, sel=sel[0], out=ab);
Mux16(a=c, b=d, sel=sel[0], out=cd);
Mux16(a=ab, b=cd, sel=sel[1], out=out);
}
b 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 dec
c 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 dec
d 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 dec
sel 0 0 dec
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
Test
Steps: 8 Outputs: 0 Slow
Fast

HDL --LBSt Project 2 ALU

```
    IN
        x[16],
        y[16],
        zx,
        nx,
        zy,
        ny,
        f,
        no;

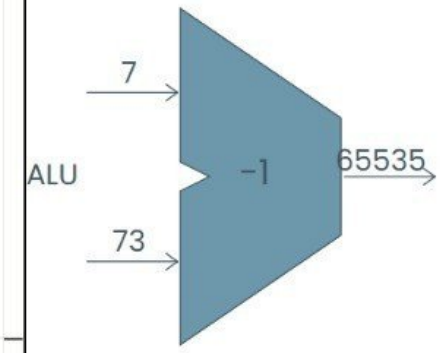
    OUT
        out[16],
        zr,
        ng;

    PARTS:
    BUILTIN ALU;
```

zr 0

ng 1

Visualization



Test Steps: 0 Outputs: 0 Slow Fast

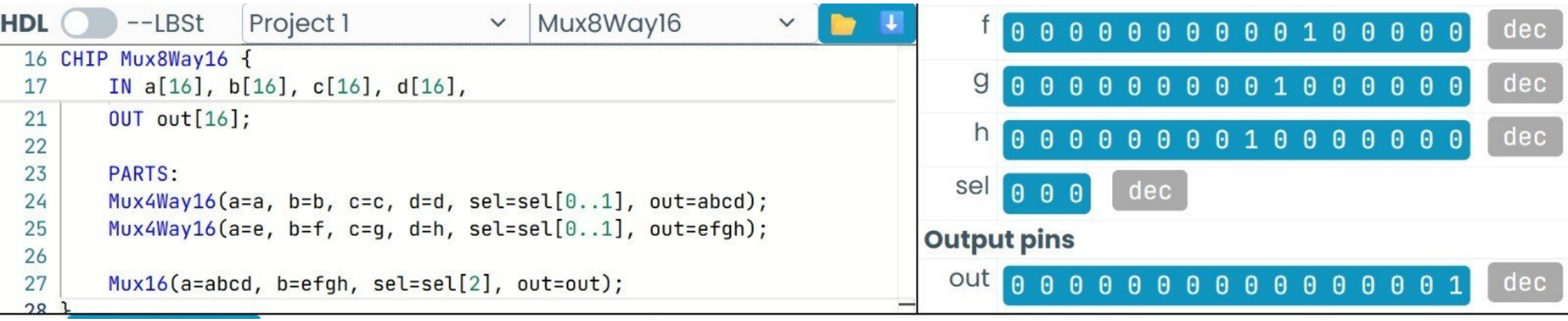
Test Script | Compare File | Output File | Diff Table

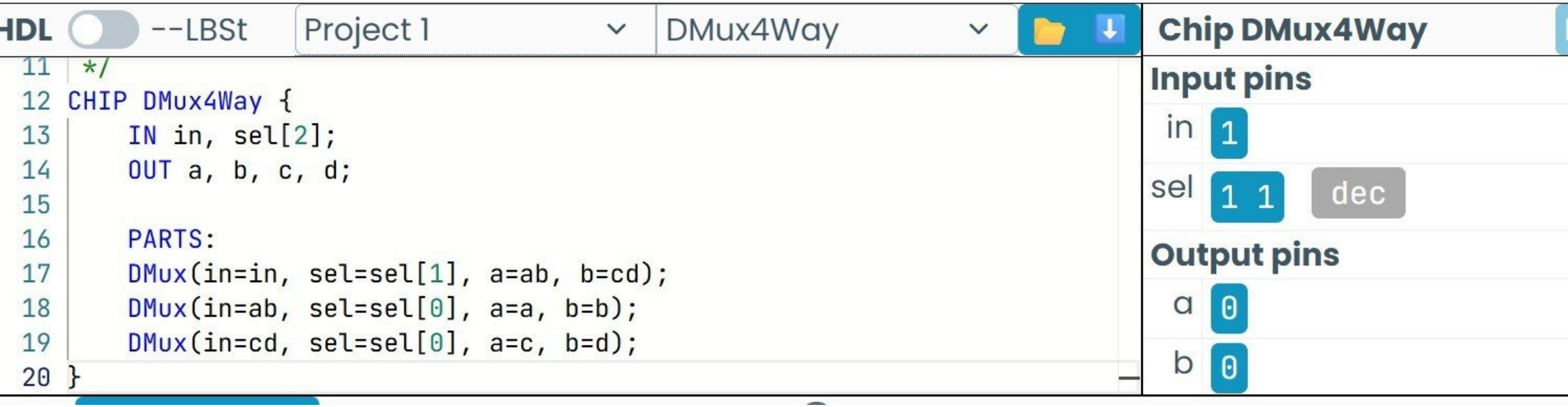
```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/ALU.tst

load ALU.hdl,
compare-to ALU.cmp,
output-list x%B1.16.1 y%B1.16.1 zx nx zy ny f no out zr ng;
```

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HDL
--LBSt
Project 1
Mux8Way16
16 CHIP Mux8Way16 {
17 IN a[16], b[16], c[16], d[16],
21 OUT out[16];
22
23 PARTS:
24 Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], out=abcd);
25 Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], out=efgh);
26
27 Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
28 }
f 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 dec
g 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 dec
h 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 dec
sel 0 0 0 dec
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec


HDL
--LBSt
Project 1
DMux4Way
11 */
12 CHIP DMux4Way {
13 IN in, sel[2];
14 OUT a, b, c, d;
15
16 PARTS:
17 DMux(in=in, sel=sel[1], a=ab, b=cd);
18 DMux(in=ab, sel=sel[0], a=a, b=b);
19 DMux(in=cd, sel=sel[0], a=c, b=d);
20 }
Chip DMux4Way
Input pins
in 1
sel 1 1 dec
Output pins
a 0
b 0

HDL --LBSt Project 1 And

```
1
2 CHIP And {
3     IN a,b;
4     OUT out;
5
6     PARTS:
7     Nand(a=a,b=b,out=nandout);
8     Nand(a=nandout,b=nandout, out=out);
9 }
```

Chip And Eval Reset Clock: 0

Input pins

a 1

b 0

Output pins

out 0

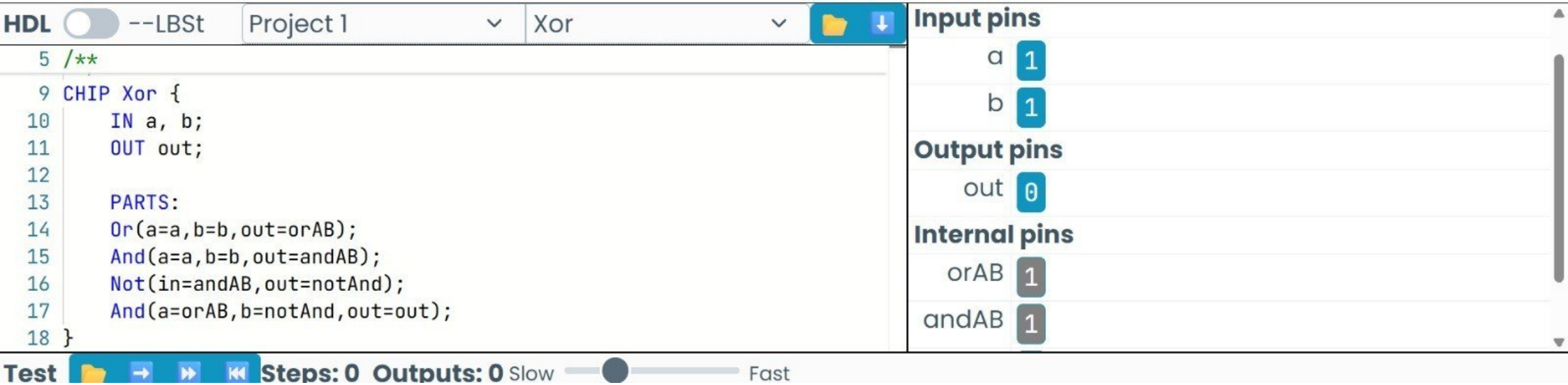
Internal pins

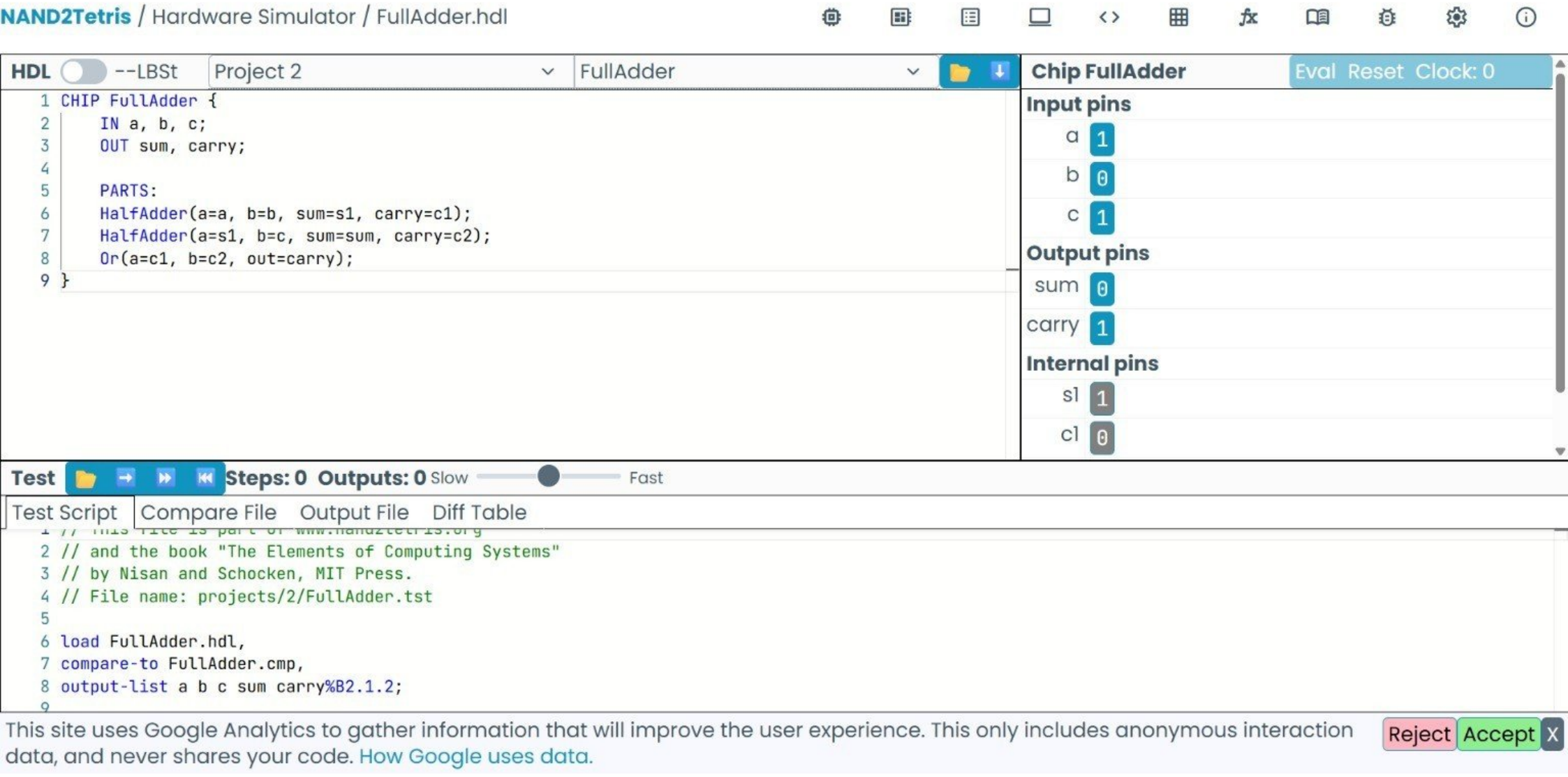
nandout 1

Test **Steps: 0 Outputs: 0** Slow Fast

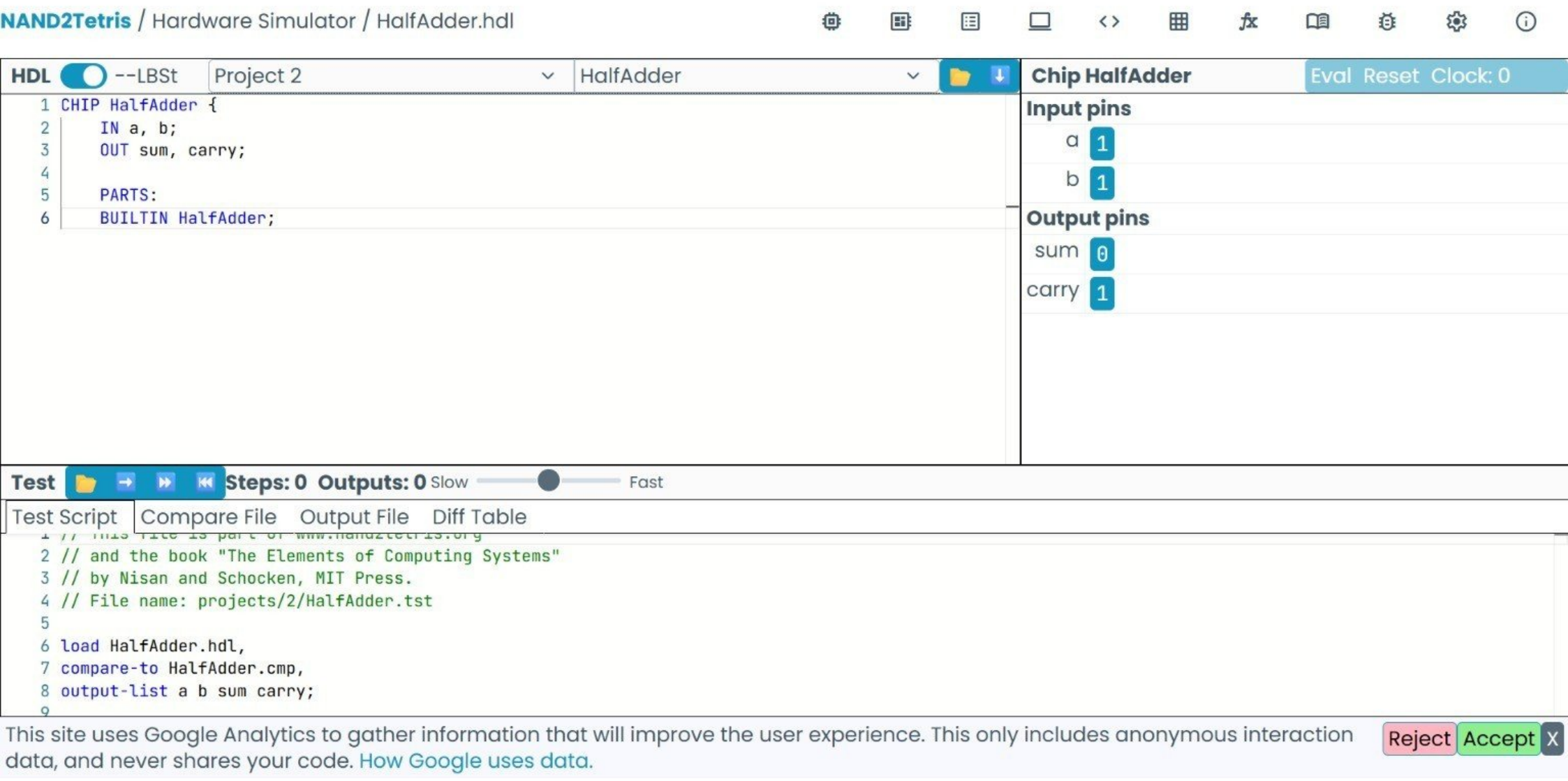
Test Script | Compare File | Output File | Diff Table

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.tst
```


HDL
--LBSt
Project 1
Xor
5 /**
9 CHIP Xor {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Or(a=a,b=b,out=orAB);
15 And(a=a,b=b,out=andAB);
16 Not(in=andAB,out=notAnd);
17 And(a=orAB,b=notAnd,out=out);
18 }
Input pins
a 1
b 1
Output pins
out 0
Internal pins
orAB 1
andAB 1
Test
Steps: 0 Outputs: 0 Slow
Fast



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HDL --LBSt Project 1 DMux

```
1 CHIP DMux {
2     IN in, sel;
3     OUT a, b;
4
5     PARTS:
6     Not(in=sel, out=nsel);
7     And(a=in, b=nsel, out=a);
8     And(a=in, b=sel, out=b);
9 }
```

Chip DMux Eval Reset Clock: 0

Input pins

in	1
sel	0

Output pins

a	1
b	0

Internal pins

nsel	1

Test **Steps: 1 Outputs: 0** Slow Fast

Test Script | Compare File | Output File | Diff Table

HDL code: No syntax errors

HDL --LBSt Project 1 Or16

Open an HDL file using this menu

```
1  CHIP Or16 {
2      IN a[16], b[16];
3      OUT out[16];
4
5      PARTS:
6      Or(a=a[0], b=b[0], out=out[0]);
7      Or(a=a[1], b=b[1], out=out[1]);
8      Or(a=a[2], b=b[2], out=out[2]);
9      Or(a=a[3], b=b[3], out=out[3]);
10     Or(a=a[4], b=b[4], out=out[4]);
11     Or(a=a[5], b=b[5], out=out[5]);
12     Or(a=a[6], b=b[6], out=out[6]);
13     Or(a=a[7], b=b[7], out=out[7]);
14     Or(a=a[8], b=b[8], out=out[8]);
15     Or(a=a[9], b=b[9], out=out[9]);
16     Or(a=a[10], b=b[10], out=out[10]);
```

Chip Or16 Eval Reset Clock: 0

Input pins

a	0 1 0 0 0 1 0 1 0 0 1 0 1 0 1 0	dec
b	1 0 1 0 1 0 0 1 0 0 1 1 0 0 1 0	dec

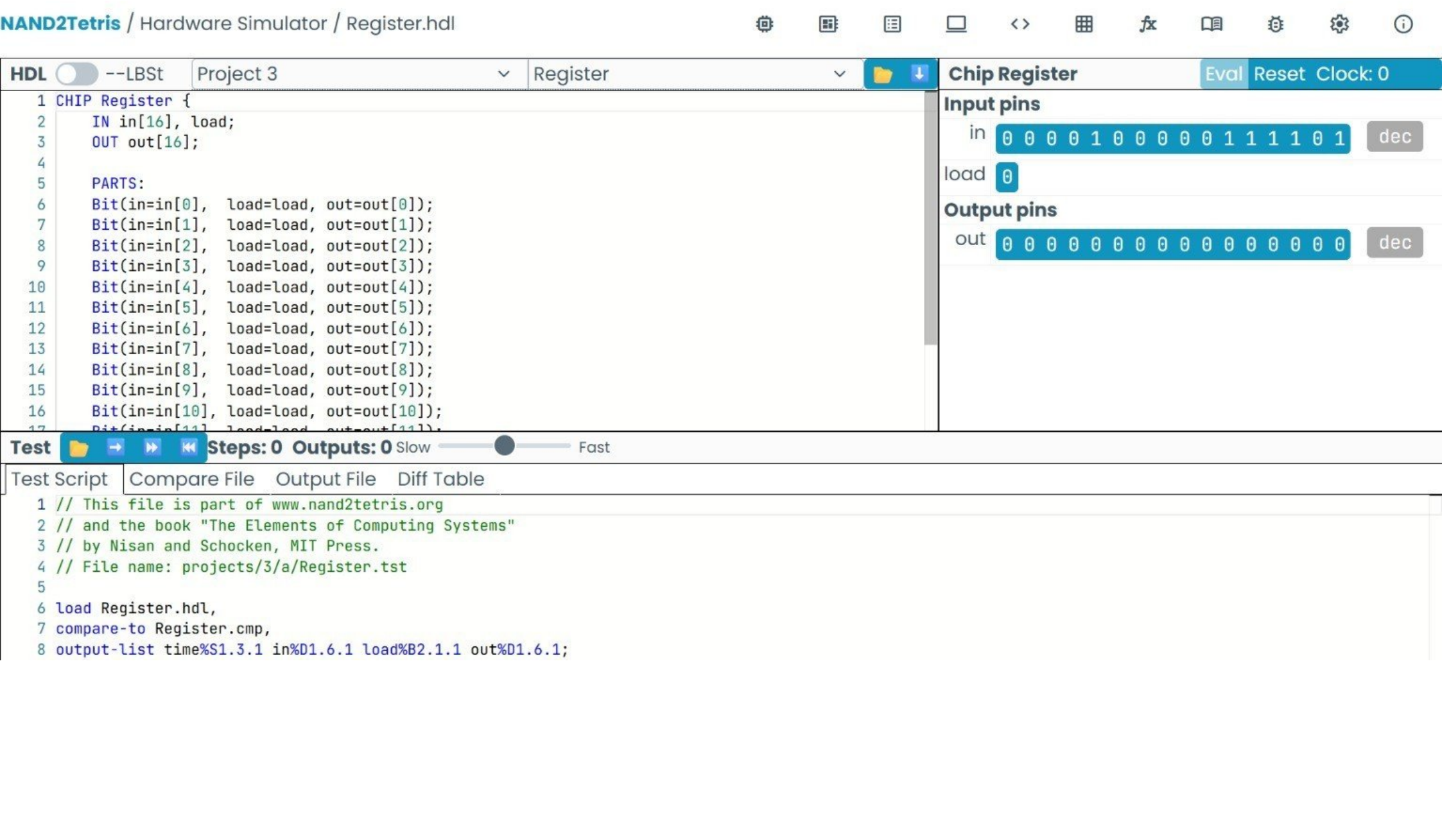
Output pins

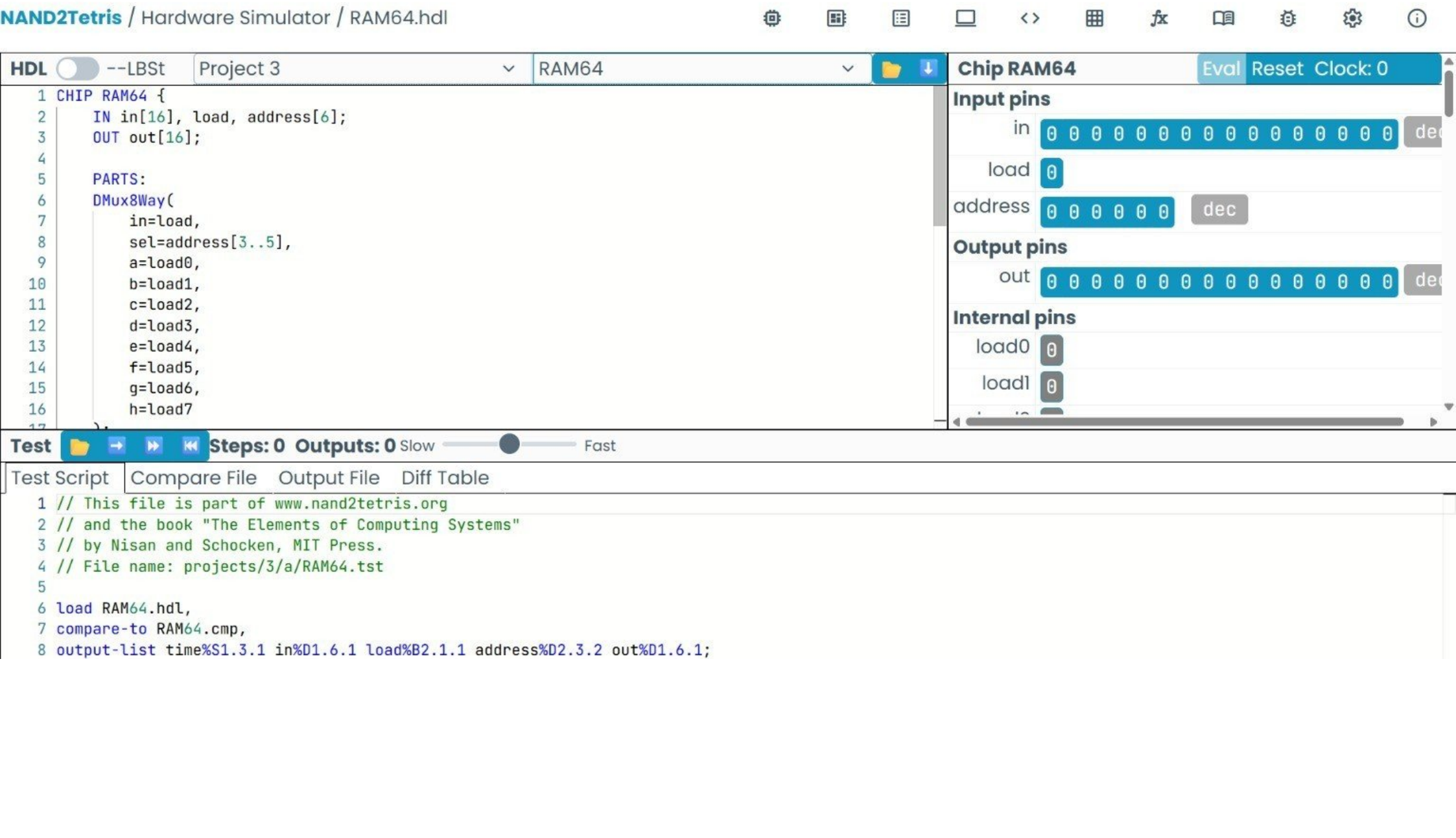
out	1 1 1 0 1 1 0 1 0 0 1 1 1 0 1 0	dec

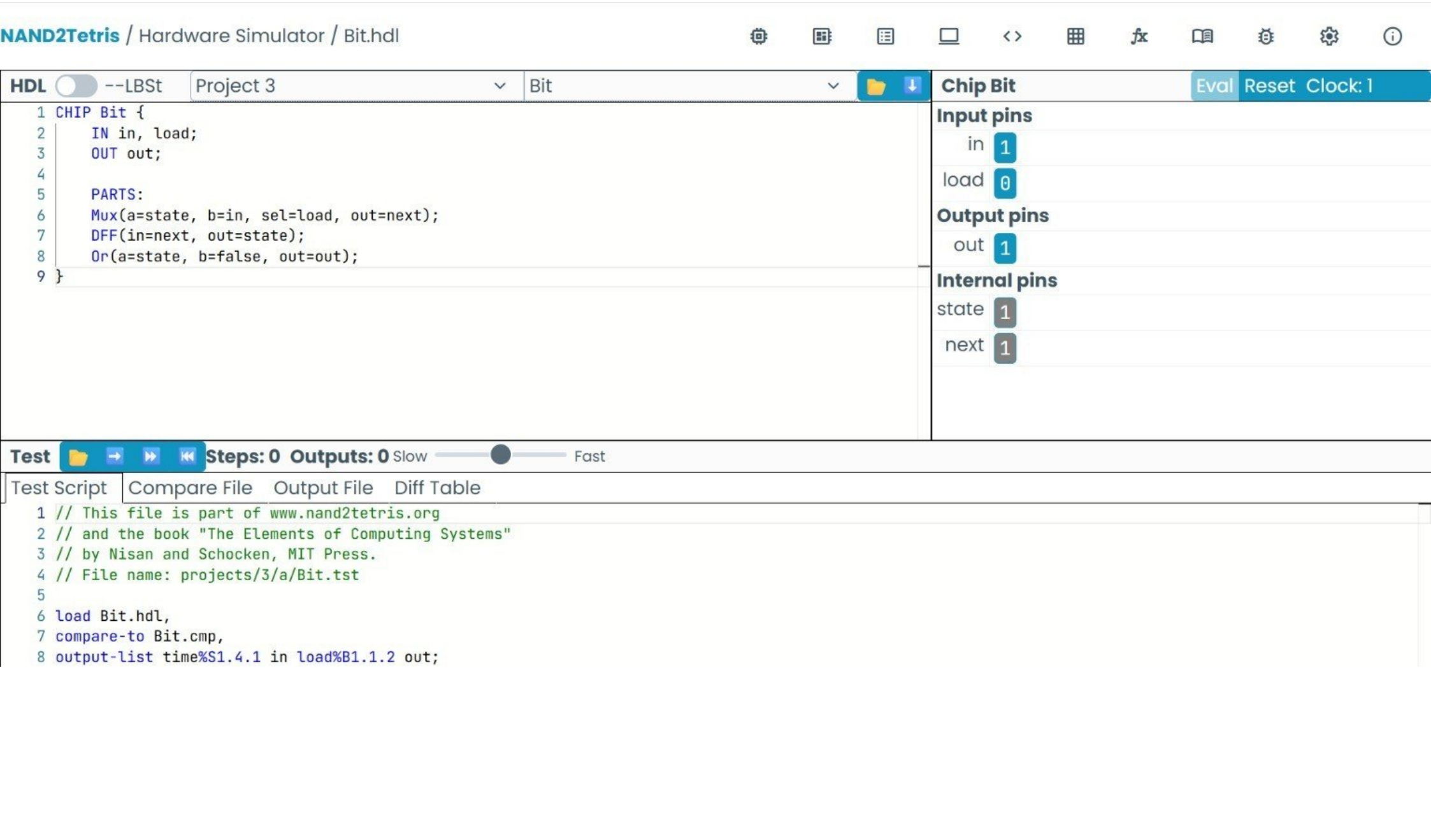
Test **Steps: 1 Outputs: 0** Slow Fast

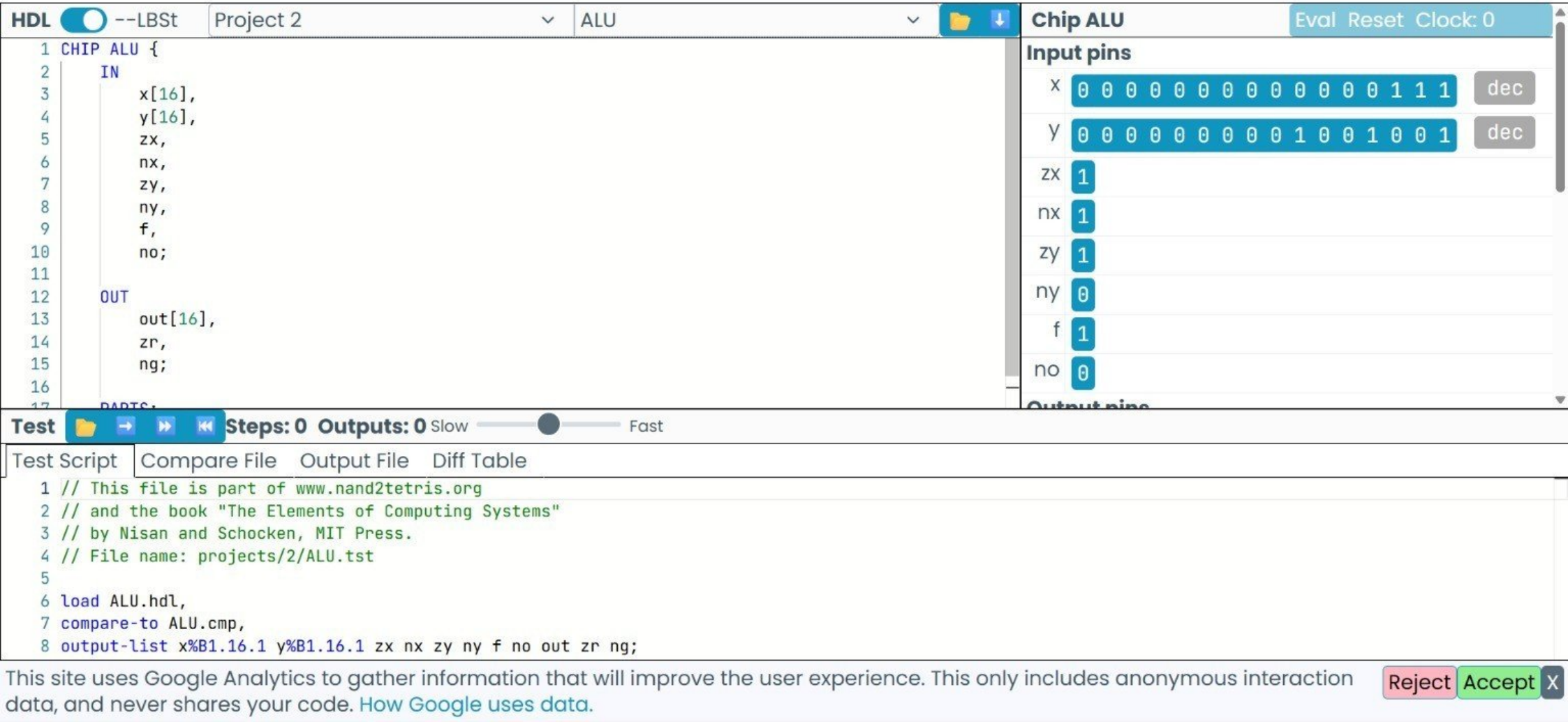
Test Script | Compare File | Output File | Diff Table

HDL code: No syntax errors


NAND2Tetris / Hardware Simulator / Register.hdl
HDL --LBSt
Project 3
Register
1 CHIP Register {
2 IN in[16], load;
3 OUT out[16];
4
5 PARTS:
6 Bit(in=in[0], load=load, out=out[0]);
7 Bit(in=in[1], load=load, out=out[1]);
8 Bit(in=in[2], load=load, out=out[2]);
9 Bit(in=in[3], load=load, out=out[3]);
10 Bit(in=in[4], load=load, out=out[4]);
11 Bit(in=in[5], load=load, out=out[5]);
12 Bit(in=in[6], load=load, out=out[6]);
13 Bit(in=in[7], load=load, out=out[7]);
14 Bit(in=in[8], load=load, out=out[8]);
15 Bit(in=in[9], load=load, out=out[9]);
16 Bit(in=in[10], load=load, out=out[10]);
Chip Register
Eval
Reset
Clock: 0
Input pins
in
0 0 0 0 1 0 0 0 0 0 1 1 1 1 0 1
dec
load
0
Output pins
out
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
dec
Test
Steps: 0 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/Register.tst
5
6 load Register.hdl,
7 compare-to Register.cmp,
8 output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 out%D1.6.1;


NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL --LBSt Project 3 RAM64
CHIP RAM64 {
IN in[16], load, address[6];
OUT out[16];
PARTS:
DMux8Way(
in=load,
sel=address[3..5],
a=load0,
b=load1,
c=load2,
d=load3,
e=load4,
f=load5,
g=load6,
h=load7
Chip RAM64 Eval Reset Clock: 0
Input pins
in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
load 0
address 0 0 0 0 0 0 dec
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Internal pins
load0 0
load1 0
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.tst
load RAM64.hdl,
compare-to RAM64.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.3.2 out%D1.6.1;


NAND2Tetris / Hardware Simulator / Bit.hdl
HDL --LBSt Project 3 Bit
1 CHIP Bit {
2 IN in, load;
3 OUT out;
4
5 PARTS:
6 Mux(a=state, b=in, sel=load, out=next);
7 DFF(in=next, out=state);
8 Or(a=state, b=false, out=out);
9 }
Chip Bit Eval Reset Clock: 1
Input pins
in 1
load 0
Output pins
out 1
Internal pins
state 1
next 1
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/Bit.tst
5
6 load Bit.hdl,
7 compare-to Bit.cmp,
8 output-list time%S1.4.1 in load%B1.1.2 out;



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NAND2Tetris
/ Hardware Simulator / And16.hdl

memory developer_board list_alt computer code grid_on function menu_book bug_report settings

HDL --LBSt Project 1 And16

```
1  CHIP And16 {
2      IN a[16], b[16];
3      OUT out[16];
4
5      PARTS:
6      And(a=a[0], b=b[0], out=out[0]);
7      And(a=a[1], b=b[1], out=out[1]);
8      And(a=a[2], b=b[2], out=out[2]);
9      And(a=a[3], b=b[3], out=out[3]);
10     And(a=a[4], b=b[4], out=out[4]);
11     And(a=a[5], b=b[5], out=out[5]);
12     And(a=a[6], b=b[6], out=out[6]);
13     And(a=a[7], b=b[7], out=out[7]);
14     And(a=a[8], b=b[8], out=out[8]);
15     And(a=a[9], b=b[9], out=out[9]);
16     And(a=a[10], b=b[10], out=out[10]);
```

Chip And16 Eval Reset Clock: 0

Input pins

a	1 0 1 0 1 0 1 1 0 1 0 1 0 1 1 1	dec
b	0 0 1 0 1 1 0 0 1 0 0 1 0 0 1 1	dec

Output pins

out	0 0 1 0 1 0 0 0 0 0 0 1 0 0 1 1	dec

Test **Steps: 1 Outputs: 0** Slow Fast

Test Script | Compare File | Output File | Diff Table

HDL code: No syntax errors